

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed full-stack, cloud-native Java Spring Boot microservices with **RESTful APIs** and modern **React** UI components, integrating **LLM/generative AI capabilities** for workflow summarization and intelligent assistant features; deployed via **Docker**, **Kubernetes**, and **CI/CD pipelines (Jenkins)** to production supporting **99.9% SLA** and **50+ concurrent users**.
- Built and operationalized **IoT device integrations** including telemetry ingestion, real-time event processing, device health monitoring, and **cloud-native IoT architectures** across **AWS (EC2, S3, Lambda, CloudWatch)**; implemented distributed event-driven patterns to aggregate and visualize **10,000+ device data streams** with sub-100ms latency.
- Engineered **LLM/generative AI productization** pipelines using **scikit-learn**, **PyTorch**, and **Hugging Face** for retrieval-augmented generation (RAG) and copilot workflows; integrated **AI/ML frameworks** with backend services to deliver contextual summarization and workflow orchestration, reducing manual processing time by **35%**.
- Developed comprehensive data pipelines and analytics using **Python (pandas, NumPy)**, **SQL (PostgreSQL, Oracle, MySQL)**, and **Power BI dashboards** with **DAX measures** and data modelling; synthesized insights from large, diverse operational datasets to drive system reliability improvements and identify hidden performance bottlenecks, reducing incident response time by **40%**.
- Executed end-to-end software solutions through system design, application development, technical troubleshooting, and production incident management using **Splunk**, **Dynatrace**, **CloudWatch** monitoring; maintained core microservices and algorithms with **secure, high-quality code** across distributed systems and **microservices architectures**, achieving **P1/P2 SLA compliance of 98%+**.
- Produced architecture and design artifacts for complex applications integrating **IoT protocols**, **device management patterns**, and **edge computing** concepts; collaborated across engineering teams and stakeholders with effective communication and proactive continuous improvement mindset; demonstrated expertise in **system design**, **application reliability**, and **security best practices** for IoT-enabled, AI-driven products.

SKILLS

Core Languages & Frameworks – Java, Spring Boot, Python, React, Angular, TypeScript, REST APIs, Flask, FastAPI.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Helm, CI/CD (Jenkins, Git), cloud-native development patterns, event-driven architectures.

AI/ML & Data – LLM/generative AI, PyTorch, TensorFlow, scikit-learn, Hugging Face, retrieval-augmented generation (RAG), workflow orchestration, pandas, NumPy, data modelling, Power BI, DAX, Tableau, statistical analysis.

IoT, Distributed Systems & Databases – IoT architectures, device management, telemetry ingestion, event processing, edge computing, distributed systems, PostgreSQL, MySQL, Oracle, relational and NoSQL databases.

Monitoring, Observability & Operations – Splunk, Dynatrace, CloudWatch, production support, incident management, SLA compliance, system reliability, operationalization, troubleshooting.

Domain & Compliance – Banking/payments systems, SWIFT messaging (MT/MX), ISO 20022, SEPA, PSD2, cross-border payments, GDPR, data protection, change-management, ERP integrations.

Data Engineering & Analytics – SQL, ETL pipelines, data warehousing, Power Query, exploratory data analysis, A/B testing, time-series analysis, Spark, distributed computing concepts.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Smart Building IoT & Generative AI Integration Platform (HCL Technologies)

- Architected and delivered a full-stack, cloud-native platform integrating **IoT device telemetry** (occupancy sensors, HVAC systems, energy meters) with **LLM/generative AI** capabilities for predictive facility management and intelligent building operations; built **Java Spring Boot microservices** backed by **PostgreSQL** and **NoSQL databases**, deployed on **AWS (EC2, Lambda, S3)** with **Kubernetes orchestration**.
- Implemented **IoT event-driven architecture** using **distributed systems patterns** to ingest and process **50,000+ real-time device events per second** with sub-100ms latency; engineered **edge computing** integrations for on-device filtering and aggregation, reducing cloud ingestion load by **45%** while maintaining full audit trails for compliance.
- Integrated **generative AI / LLM capabilities (Hugging Face, PyTorch)** for automated facility summarization, anomaly detection, and copilot-driven maintenance workflows; built **RAG-style retrieval pipelines** using **Python** to contextualize device data with historical facility records, enabling **95% accuracy** in predictive maintenance recommendations.
- Developed **Power BI dashboards** with **DAX measures** and advanced data modelling to visualize **IoT sensor trends, device health**, and operational KPIs; synthesized insights from **large, diverse datasets** to identify hidden performance patterns and drive **20% energy efficiency improvements**.
- Designed and executed **system reliability and security best practices** for IoT-enabled products, including device authentication, encrypted payload transport, and **cloud-native security patterns**; deployed via **Docker, Kubernetes, Helm** and **CI/CD tooling (Jenkins)** to achieve **99.95% availability** across production workloads serving **100+ enterprise buildings**.

Real-Time Payments Processing & Device Status Intelligence Dashboard (HCL Technologies)

- Built production-grade operational intelligence platform for banking payments processing, integrating **SWIFT messaging (MT/MX formats)**, **ISO 20022 message flows**, and cross-system **REST API integrations** across **Oracle, PostgreSQL** databases; delivered **Java Spring Boot backend** with **React frontend**, deployed on **AWS** with **CloudWatch monitoring**.
- Engineered **IoT-style device status and health monitoring** for payment node infrastructure, ingesting telemetry from **distributed transaction processing systems**; used **Python ETL pipelines** and **Splunk, Dynatrace observability** to detect operational anomalies in **real-time**, reducing Mean Time to Detection (MTTD) by **55%**.
- Synthesized **large, diverse datasets** from payment transaction logs, regulatory audit trails, and system performance metrics; built **Power BI dashboards** with **DAX calculations** to provide **GDPR-compliant** reporting on payment flows, transaction status, and regulatory compliance, supporting **10+ stakeholder teams** and **95%+ SLA adherence**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)